



06: Conceptualising & Contextualising

IN3065/INM355 | 2025–2026

User-Centred System Design

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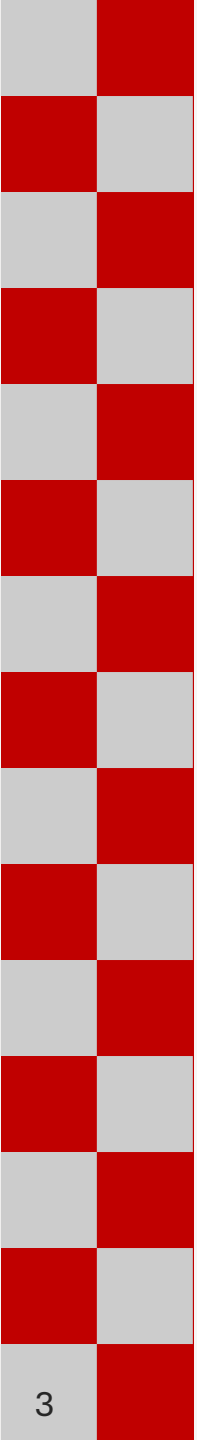
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Session 6: Learning objectives



By completing this session, including any practical activity and assigned readings, you should be able to:

- Use Crazy 8s to generate multiple design ideas quickly
- Describe techniques for evaluating early design concepts
- Apply approaches for contextualising design ideas in realistic use situations, including
 - Scenarios
 - Storyboards
 - Future User Journey Maps
- Explain what mental models are and why designs should align with users' mental models



Conceptual design & ideation continued

Recap: Conceptual design

- Conceptual design is **high-level** design work.
- It **bridges** the *problem space* and the *design space*
- It focuses on **what kind of product or system** it is and **what facilities it must offer**, not the interface details.
- Begin by generating **lots of design ideas**.
- Ideas are **evaluated and refined**, with weaker ones discarded.
- Design is **iterative**—exploration and elimination of alternatives is repeated.
- You seldom get it right the first time—the first idea is rarely the best, so give it competition!

Recap:

Sketching for ideation: Crazy 8s

What is it?

- A core Design Sprint method, part of the [Sketch](#) stage.
 - (usually preceded by warm-up / boot-up activities)
- Rapid sketching exercise: 8 rough sketches in 8 minutes
- A technique for generating ideas.

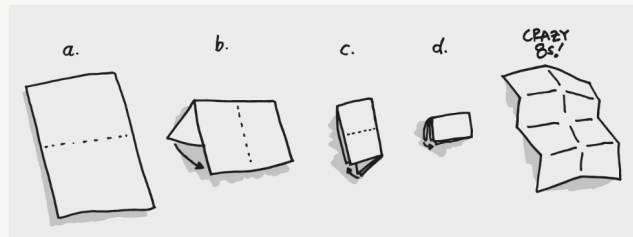
Why do it?

- Generates lots of ideas, at a rapid pace
 - First ideas typically aren't the best—this technique forces you to think beyond those.
 - 8 minutes (1 minute per sketch) isn't enough time for people to get bogged down in detail or perfection.
- <https://designsprintkit.withgoogle.com/methodology/phase3-sketch/crazy-8s>

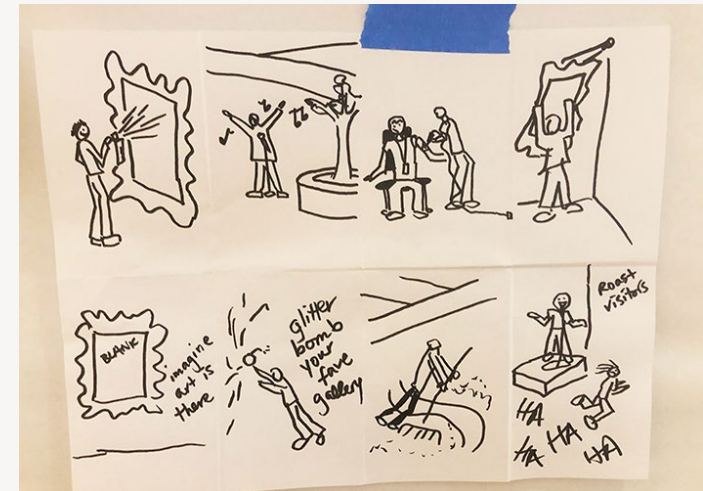
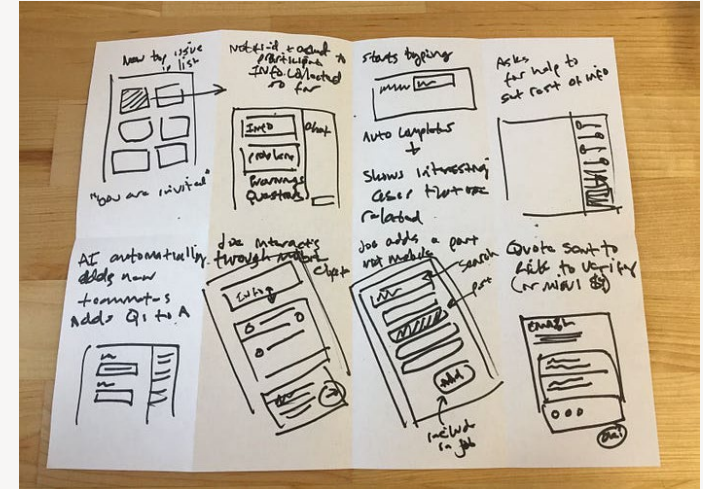


Crazy 8s: **How** to do it

1. Each person has a piece of paper with 8 squares (fold a page into 8 sections, or use a printed template)



2. Set the timer for eight minutes
3. Each person sketches one idea in each section of their paper, until they have filled all 8 sections.
4. When the timer goes off, put pens down.



Images from:

<https://uxdesign.cc/honing-crazy-eights-with-randomness-1c8100b94824>

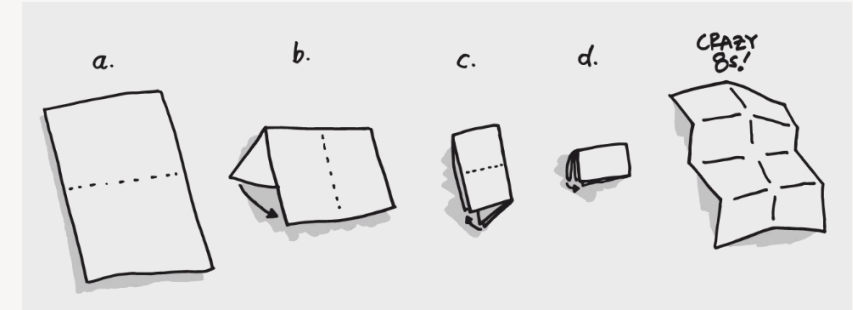
<https://designthinkingformuseums.net/2019/07/27/bad-ideas-brainstorming>

Crazy 8s: **What next?**

- It depends on what you're trying to achieve, or the process you're following.
- Typically, ideas are first **reviewed**, to decide which to explore/develop further.
- When you've **shortlisted** ideas, you might try to **progress** them.
 - E.g. coming up with more detail, adapting them, envisaging scenarios for them, creating storyboards, more sketches, future user journey maps....
it depends on what you'll find useful. Experiment!
- OR you might do **another round** of Crazy 8s, using the top/shortlisted ideas as a starting point / inspiration

Exercise:

Crazy 8s for **concept ideation**



The design brief:

People across London spend significant time waiting at bus stops, often in environments that feel boring, uncomfortable, or uncertain.

Your task is to come up with **high-level concept ideas**, informed by a persona (Leila Ahmed), that could improve the experience of waiting at a bus stop.

Instructions

1. Each take a piece of paper, and fold it so it has 8 sections.
2. You have 5 minutes to read the design brief (above) and persona.
3. When the timer starts, sketch 8 ideas—one in each square.
4. When the timer ends, put down your pen.



Leila Ahmed “I’d like waiting to feel less ‘empty’”

About Leila

Leila commutes by bus most days because she likes staying on the surface—“you see more of the city.” She hops between cafés and client meetings, so she waits for buses often. She describes these little pauses as the duller part of her day.

She’s sociable but introverted; she likes feeling part of the world around her without having to speak to strangers. She’s creative, curious, and a bit of a daydreamer.

Age: 29

Lives: Stoke Newington

Occupation: Graphic designer at a small studio in Dalston

Frequent Routes: 73, 149, 243

How Might We...

- HMW help Leila feel engaged during short waits without demanding her full attention?
- HMW give Leila clear, trustworthy cues about her waiting time so she can settle?
- HMW make the bus stop feel visually richer or more atmospheric to reduce monotony?
- HMW make the bus stop feel safer and more reassuring at quieter times of day?

Goals

- Feel engaged without needing to focus deeply
- Know how long she’ll be waiting so she can relax
- Be in an environment that feels warm, safe, or visually inspiring
- Avoid falling into boring, unintentional phone use

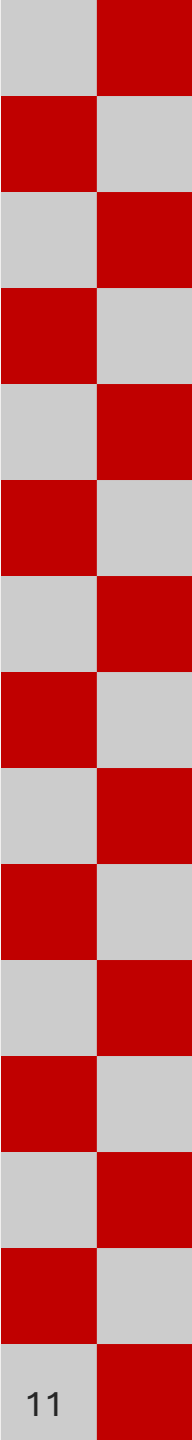
Frustrations

- Bus times that fluctuate or seem unreliable
- Monotonous or unattractive stops
- Harsh lighting that makes the space feel cold
- Other people’s stress or impatience changing the vibe
- Feeling like she’s wasting time

Habits & Behaviours

- Often stands rather than sits, unless the seat looks appealing.
- Pulls out her phone automatically but gets bored of scrolling.
- Looks around a lot to make the wait feel faster.
- Gets restless in plain or sterile-feeling stops.
- Perks up when there’s something visually interesting or atmospheric nearby.

CRAZY 8'S

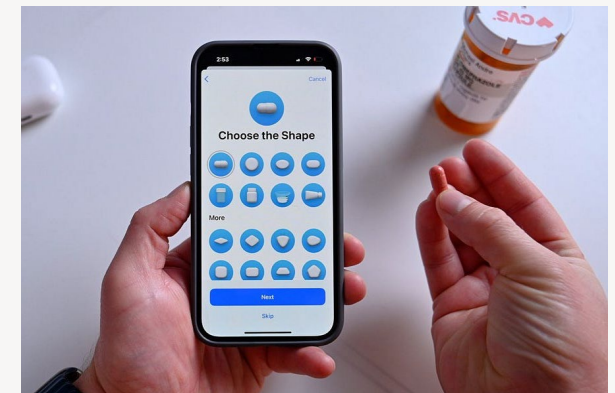
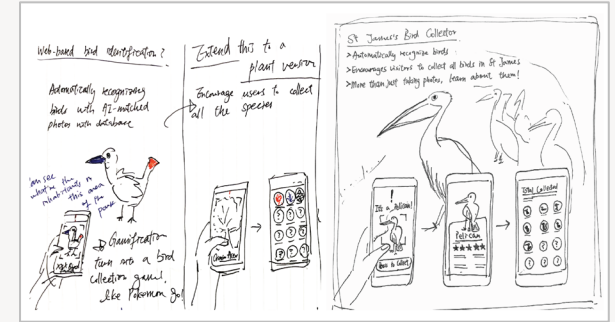
A vertical decorative element on the left side of the slide, consisting of a series of alternating red and white squares.

Evaluation 1:

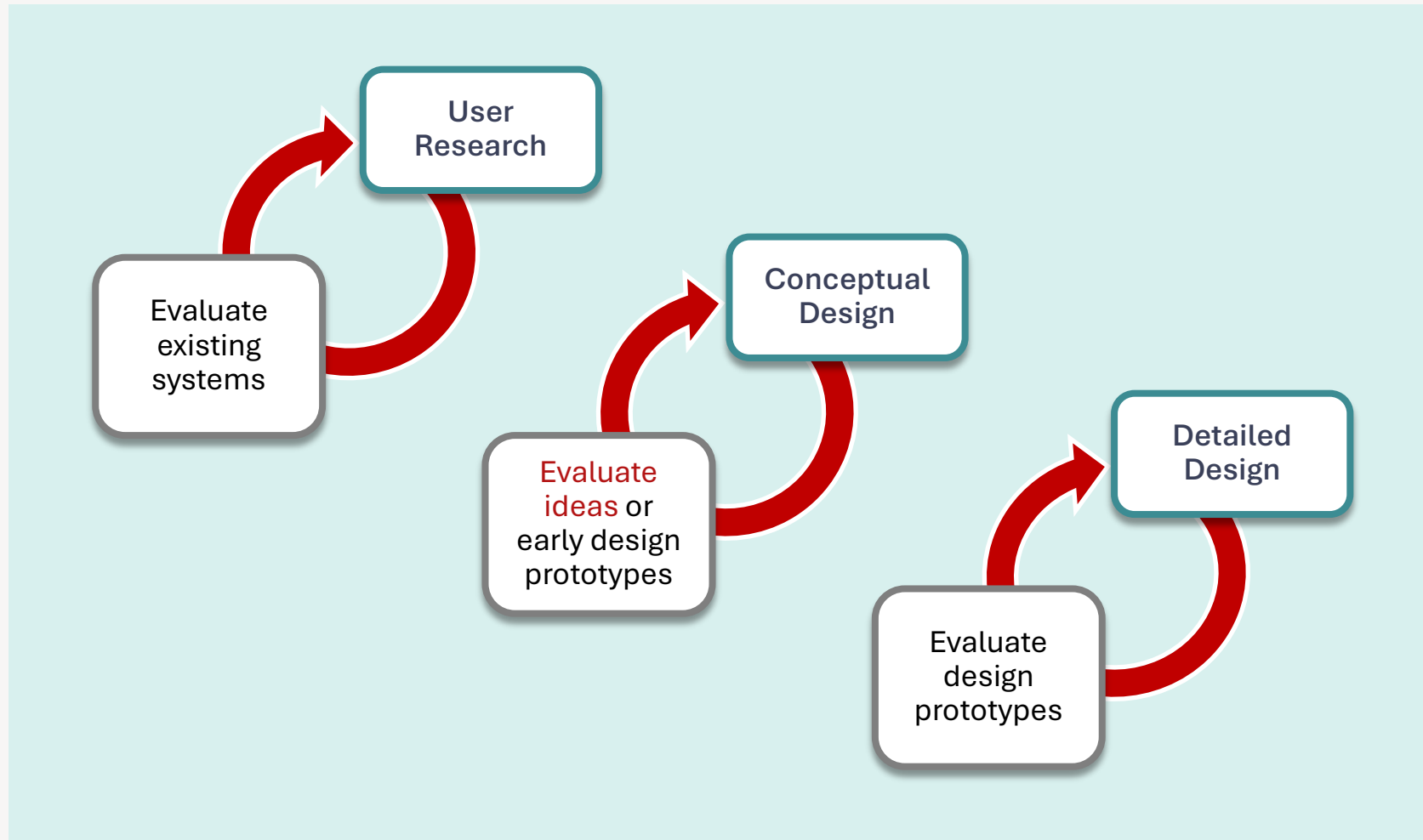
Evaluating early design ideas

Evaluating design work

- How do we know if our designs are useful?
- How do we choose between alternative ideas?
- How do we know if our designs have good usability?
- How do we know if they provide a positive user experience?
- How do we know if we need to make changes and what changes might need to be made?
- How can we iterate in our design work?



Evaluation in a user-centred design process

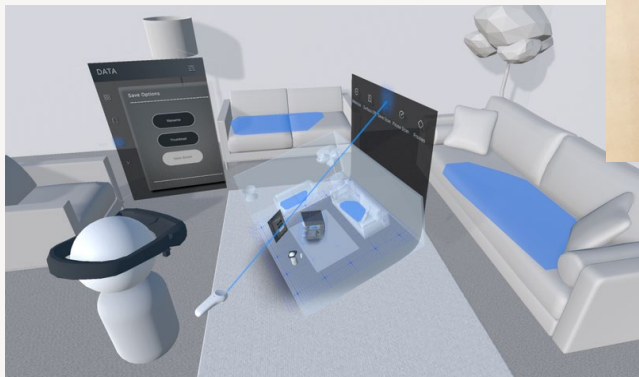
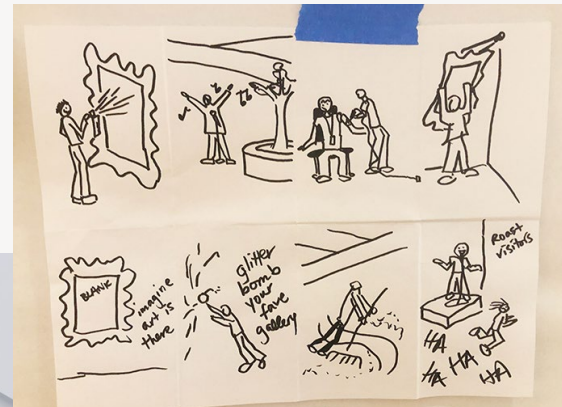
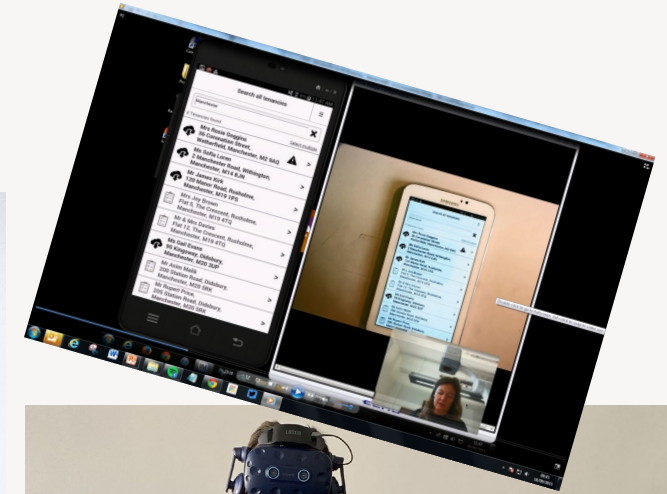


For what **purposes** do we evaluate?

- To get early feedback on design concepts.
- To decide between design options.
- To check that users can use the product and whether they like it.
- To identify and fix usability problems in designs *before* production, or release.
- To identify and fix usability problems in our existing products.
- To assess competitor systems.
- To focus on real problems—not our intuition about what is good or bad.

What can we evaluate?

- Sketches
- Paper prototypes
- Clickable digital prototypes
- Physical prototypes
- Wireframes
- Existing systems
- Competitor systems



How can we evaluate?

- Usability testing.
- Expert evaluations e.g. Heuristic Evaluation.
- Interviews
- Usability/UX questionnaires.
- Design reviews and design critique.





Evaluation in early/conceptual design

Approaches

Evaluating ideas

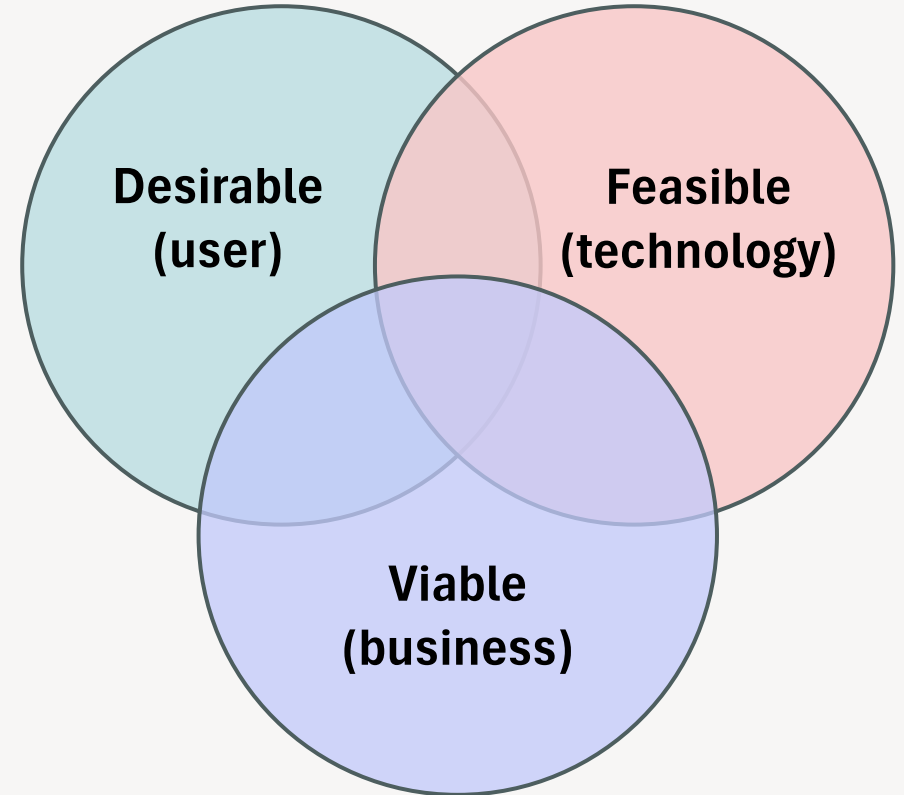
- How do you decide which ideas to progress?
- E.g. from Crazy 8s:
 1. Each person presents their top 2 or 3 ideas back to the group
 2. Vote on the best, e.g. using ‘dot voting’ (sticky dots)
 3. See which ideas are most popular.
- Then do another round of Crazy 8s or progress the ideas into more detailed designs

Product Triad: Desirable, Viable, Feasible

- **Desirable (user)**: useful and usable; alignment with user needs and desires.
- **Feasible (technology)**: budget, timeline, technical constraints.
- **Viability (business)**: financially sustainable in the longer term.

<https://www.nngroup.com/articles/the-product-triad-designs-role/>

<https://www.uxdesigninstitute.com/blog/desirability-viability-and-feasibility/>



Concept **testing**

- Present an idea, prototype, or concept to users to gather feedback
- Test concepts with users to validate (or abandon) ideas.
- Get feedback on perceived value, usability flaws.
- Use approaches such as surveys, focus groups, interviews.
- This is not about testing for usability issues.
- See: <https://www.looppanel.com/blog/ux-research-concept-testing>

Design Critique

- A collaborative approach to reviewing ideas and designs.
- Can be undertaken throughout the design process; valuable in early design.
- How?
 - Presenter shares a design.
 - Critics give constructive feedback.
 - Facilitator manages session.

See also:

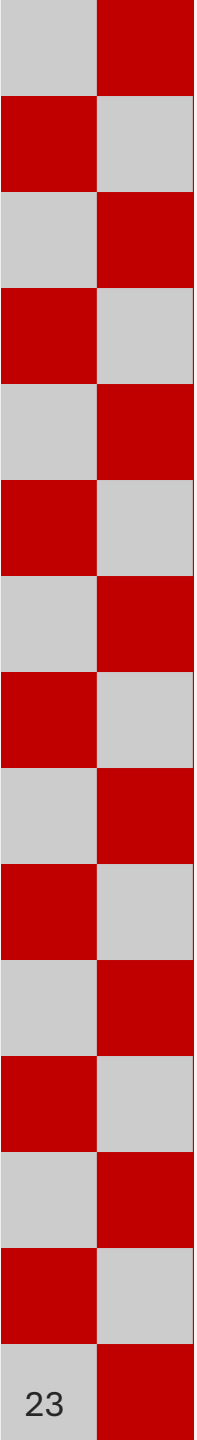
- <https://www.nngroup.com/articles/design-critiques>
- <https://www.interaction-design.org/literature/topics/design-critiques>
- <https://jmspool.medium.com/what-goes-into-a-well-done-critique-3a8780884055>



NNGROUP.COM NN/g

Questions?





Conceptual design

Contextualising

Conceptual design as **envisionment**

- Envisioning a world in which the new interactive system or experience exists.
- Concerned with the “big picture”
- Focusing on:
 - **Overall paradigm** or **platform** (e.g. web, mobile, AR/VR, voice etc).
 - **Main concepts, objects** and **actions** within the system.
 - **Key activities** and **workflows** that will be supported
 - The **division of responsibility** between user and system.
- This often involves **contextualising** out new system / designs



Contextualising concepts

- Conceptual design can involve **imagining use in context**
 - How might people interact with the system in the real world?
 - How does interaction unfold over time?
- Designers often describe **situations of use**
- Techniques for this include:
 - Scenarios
 - Storyboards
 - Future user journeys

Contextualising interaction

Imagining how a design will be used in the real world

User	Who is involved / using the system?
Context	Where and when does the interaction happen?
Activity	What are they trying to achieve?
Interaction / Journey	How does the interaction unfold over time?



Contextualising: **Scenarios**

Scenarios

- A scenario is an “*informal narrative description*” of a user’s task - one typical story (John M. Carroll, 2000).
- Captures a task in user’s language.
- An informal, personalised, context-rich account of users, events, artefacts and environments.
- Typically focuses on one persona.
- Either current situations or (more likely) envisioned future situations.
- Summarises user journeys; can have a collection of scenarios that encapsulate entire interaction with system.
- Contain important concepts and actions.

Scenarios

“Scenarios are stories. They consist of a setting, or situation state, one or more actors with personal motivations, knowledge, and capabilities, and various tools and objects that the actors encounter and manipulate.

The scenario describes a sequence of actions and events that lead to an outcome. These actions and events are related in a usage context that includes the goals, plans, and reactions of the people taking part in the episode.”

Mary Beth Rosson and John M. Carroll (2012), ‘Scenario Based Design’, in *The Human-Computer Interaction Handbook*. CRC Press. ([pdf](#) if you want to read more)

Scenarios: **contextualising** use

- Should **situate** the use of the system
- Should capture the **user needs and goals**.
- Should not dwell on unnecessary details of how the system will work.
 - Although, some representation is often necessary (e.g. when the user searches a database some sort of technology is needed!).
- Key focus should be **how the system supports users in context**.

Scenario example: ‘ShareWear’

Dev’s phone buzzes with a reminder about their best friend’s formal birthday party, which is two weeks away.

As someone really into sustainable living, Dev is reluctant to buy a new outfit they’ll wear just once. They remember a clothing share app—ShareWear—they recently heard about. They find and install it easily.

After creating an account, Dev goes straight to formalwear. There’s a lot available so they narrow in by size, but also colour, and style. A fabulous outfit, paired with matching accessories catches their eye—they decide to try it on virtually. They soon see themselves onscreen, overlaid with the outfit and accessories—it looks great. The hire cost is reasonable and when Dev checks, there are good reviews from previous customers as well, including about the outfit owner.

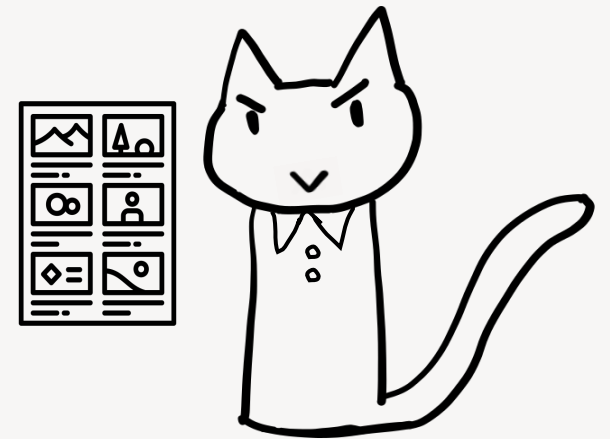
Convinced, Dev adds the outfit to their shopping basket, specifying the number of hire days. They head for the checkout, where they review their choices and decide to keep the outfit for another week. They proceed to check out, enter their delivery and payment details, and place the order. They can see in the app that the order is now processing, and a confirmation email arrives almost immediately.

Two days later, the outfit arrives. Dev tries it on, and it looks even better than they’d hoped. They feel a sense of satisfaction—not just for finding a great outfit for the party, but for doing so in a more sustainable way.

Questions?



Contextualising: Storyboards



Storyboards

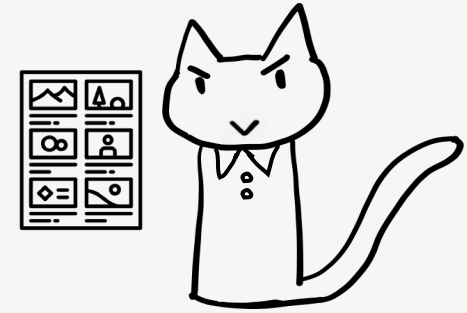


- An alternative representation of a user journey.
- Usually focuses on the user experience, not details of the design.
- A storyboard contains:
 - A sequence of panels, or cells, each representing a single step in the user's journey.
 - Captions to explain the content of each panel.
 - Speech / thought bubbles to show users' thoughts / motivations.
 - The whole story should have a clear 'beginning' and 'end'.
- Artistic quality is not important!
- <https://www.smashingmagazine.com/2017/10/storyboarding-ux-design>



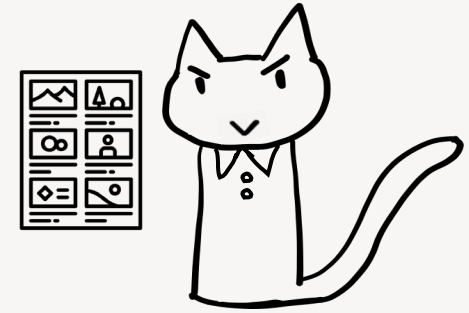
From: <https://www.smashingmagazine.com/2017/10/storyboarding-ux-design/>

Storyboards in conceptual design



- Similar to scenarios but in visual form.
 - Very compact way of showing interaction in context.
 - Highly creative.
 - Good for communicating key aspects of system.
- Very similar to storyboarding in film.
- A series of panels which show the major interaction moments with the system
- Show important **concepts**, **actions** and **context of use**.
- Use to expose proposed features of the system, showing their use in context.
- Usually hand-drawn but there are some tools: <http://www.storyboardthat.com/>

Developing a Storyboard



- Again, start with a particular persona who has a particular goal.
 - This gives the overall purpose of the story.
 - Different personas will behave differently.
- Work out the scenario or journey.
 - Decide on a starting point, and an ending point.
 - Work out key points—steps—to illustrate between them.
 - Decide on the point of view for each step/frame—what will be shown?
- Create a panel with a caption and dialogue / thought bubbles for each step.
- Try a storyboard for one persona, then see if the story varies for others.
- Explore different journeys and different digital touchpoints.



Adeola steps in after a long day at work...



...and finds her siblings hogging the TV, as usual.

She heads upstairs to study...



ugh, I can barely even see my lecture...

if i wanted to watch lectures on my tv how would i go about that?



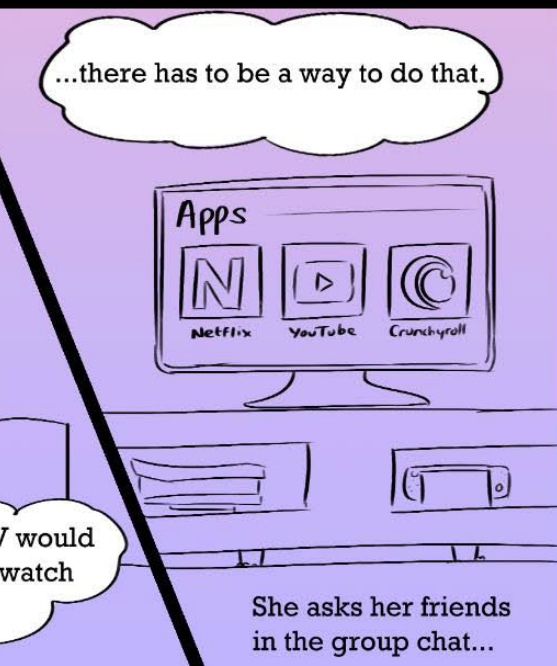
...my phone screen is just way too small. smh.

Time passes, and Adeola finishes studying.



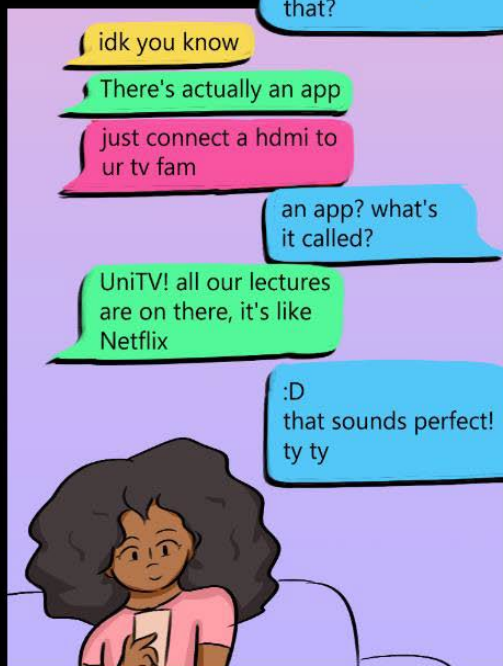
Her siblings are asleep and she finally has the TV to herself.

honestly, a TV would be perfect to watch lectures on...



...there has to be a way to do that.

She asks her friends in the group chat...



idk you know

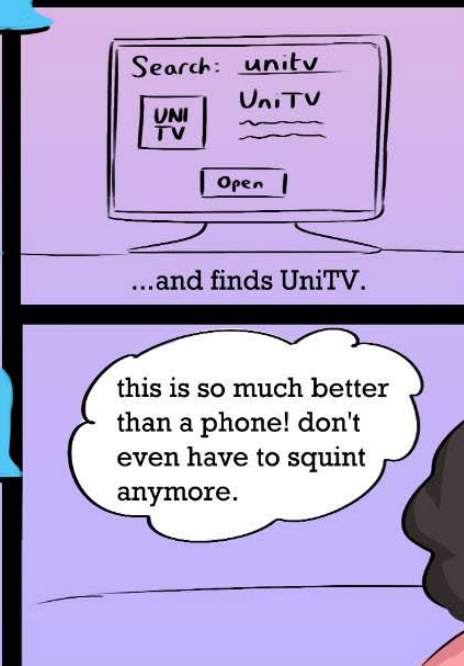
There's actually an app

just connect a hdmi to ur tv fam

an app? what's it called?

UniTV! all our lectures are on there, it's like Netflix

:D that sounds perfect! ty ty



Search: unitv
UniTV
Open

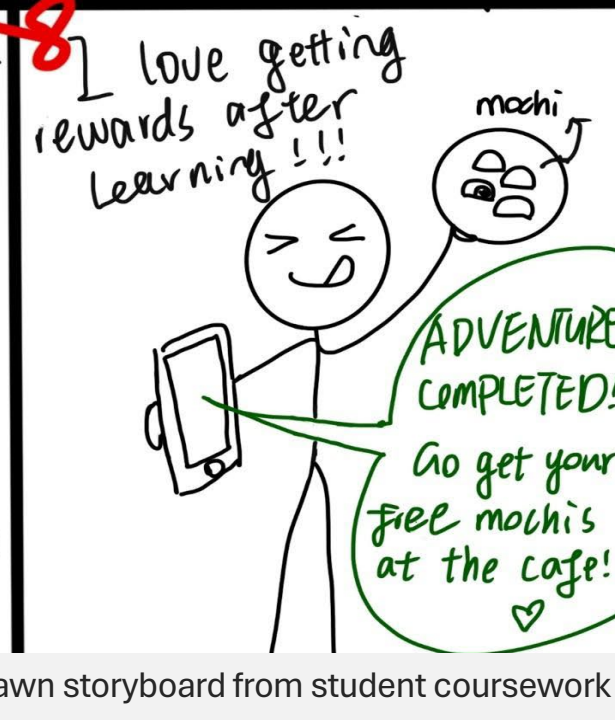
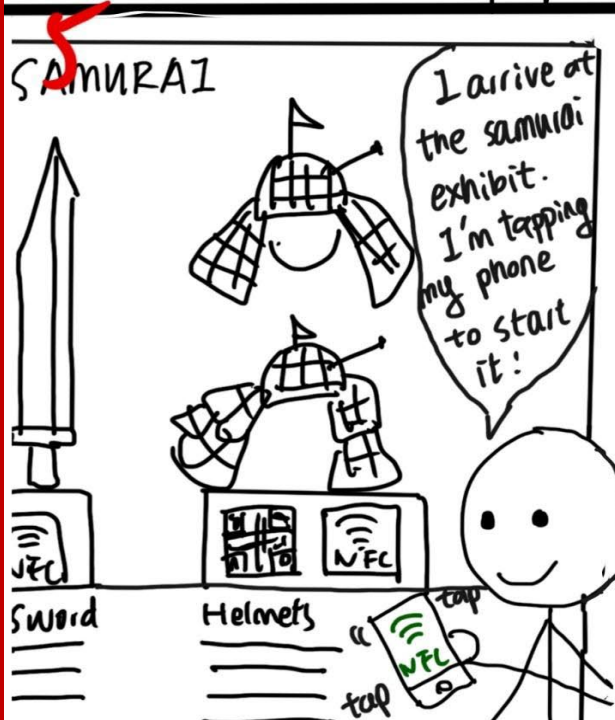
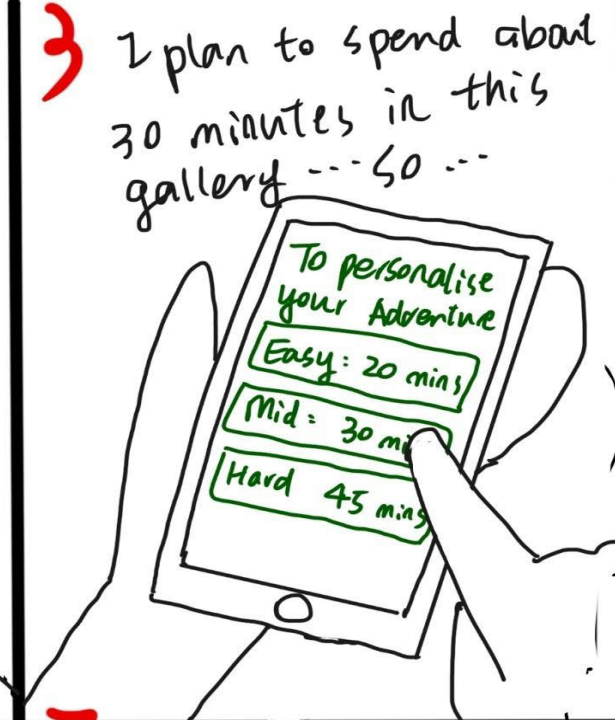
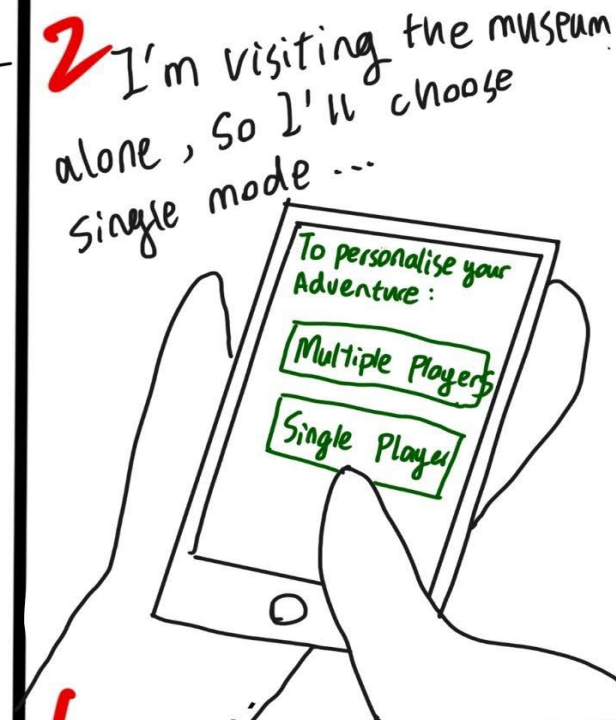
...and finds UniTV.



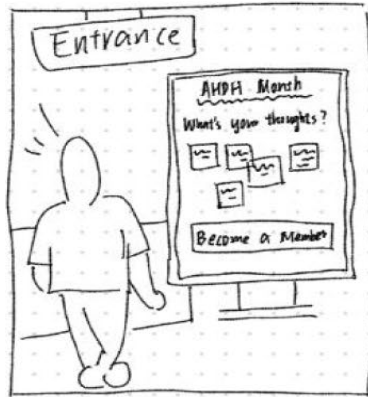
this is so much better than a phone! don't even have to squint anymore.



Hand-drawn storyboard



Experience Wellcome Library with AR Assistance App



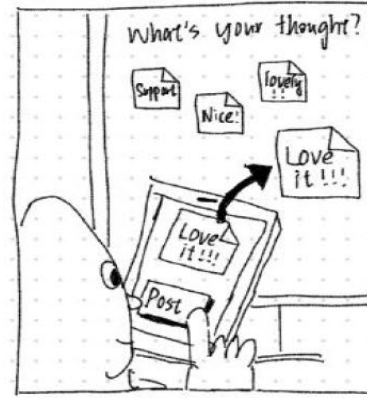
Arriving at the Library

Visitor arrives at the library entrance and notices a large digital board displaying monthly event.



Joining Library Membership

Visitor uses the AR app to scan a QR code displayed on the digital board to access membership sign-up and community features, inputting personal details for registration.



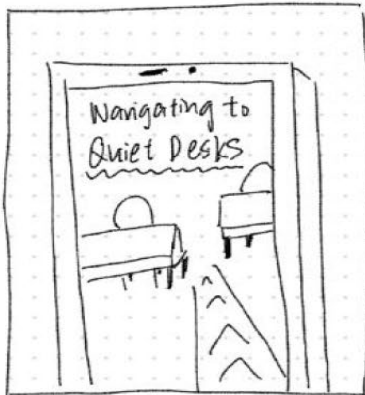
Community Engagement

Once registered, the digital board displays the current month's theme, inviting visitors to share thoughts via the AR app by responding to the theme.



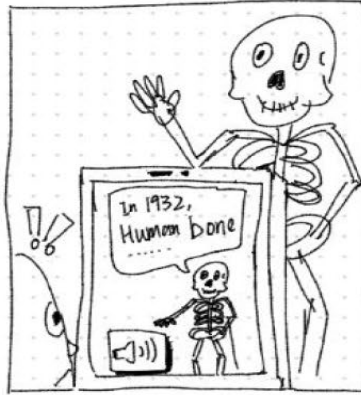
Navigation to Resources/Books

Using the AR app, the visitor selects "Resources/Books" and receives navigational prompts leading them to the section they're interested in, guided by augmented reality indicators



Finding Quiet Study Spots

Choosing "Quiet/Relaxing Study Spots" on the AR app, the visitor is guided to serene study areas within the library, indicated by virtual markers.



Exploring Exhibitions/Displays

Opting for "Exhibitions/Displays," the visitor follows AR directions to various displays, receiving additional information and interactive features through the app.



Leaving Feedback

Before departure, the AR app allows the visitor to post opinions or feedback on the digital board, sharing their thoughts on the library experience.



Re-engagement

After leaving the library, the visitor receives inbox messages notifying new exhibitions and happenings.

Tom and his son decide to visit the IWM



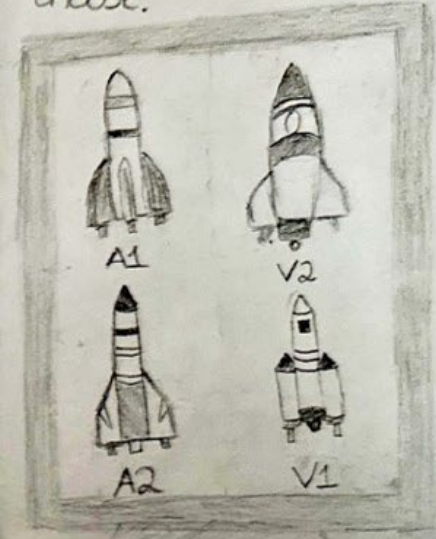
They are so impressed by the V2 rocket exhibit.



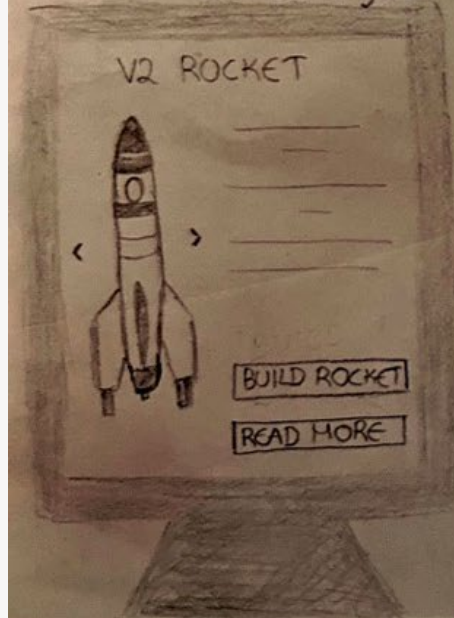
They notice an interactive kiosk near the rocket.



There are different models for them to choose.



Wow so much information about the rocket! And you can spin it around to observe all angles!



They construct the rocket and learn more about its engineering.



They experience the launch through vibrations, sound and a light breeze. That's so engaging!



They choose to 3D print the rocket to take it home and remember the experience.



~THE END~

Questions?





Contextualising: **Future User Journey Maps**

Future User Journey Maps

Similar to the user journey maps we learned about in lecture 4...

...BUT instead of describing the *current* experience discovered through research, they describe a **possible future experience enabled by a new design**.

Future journey maps:

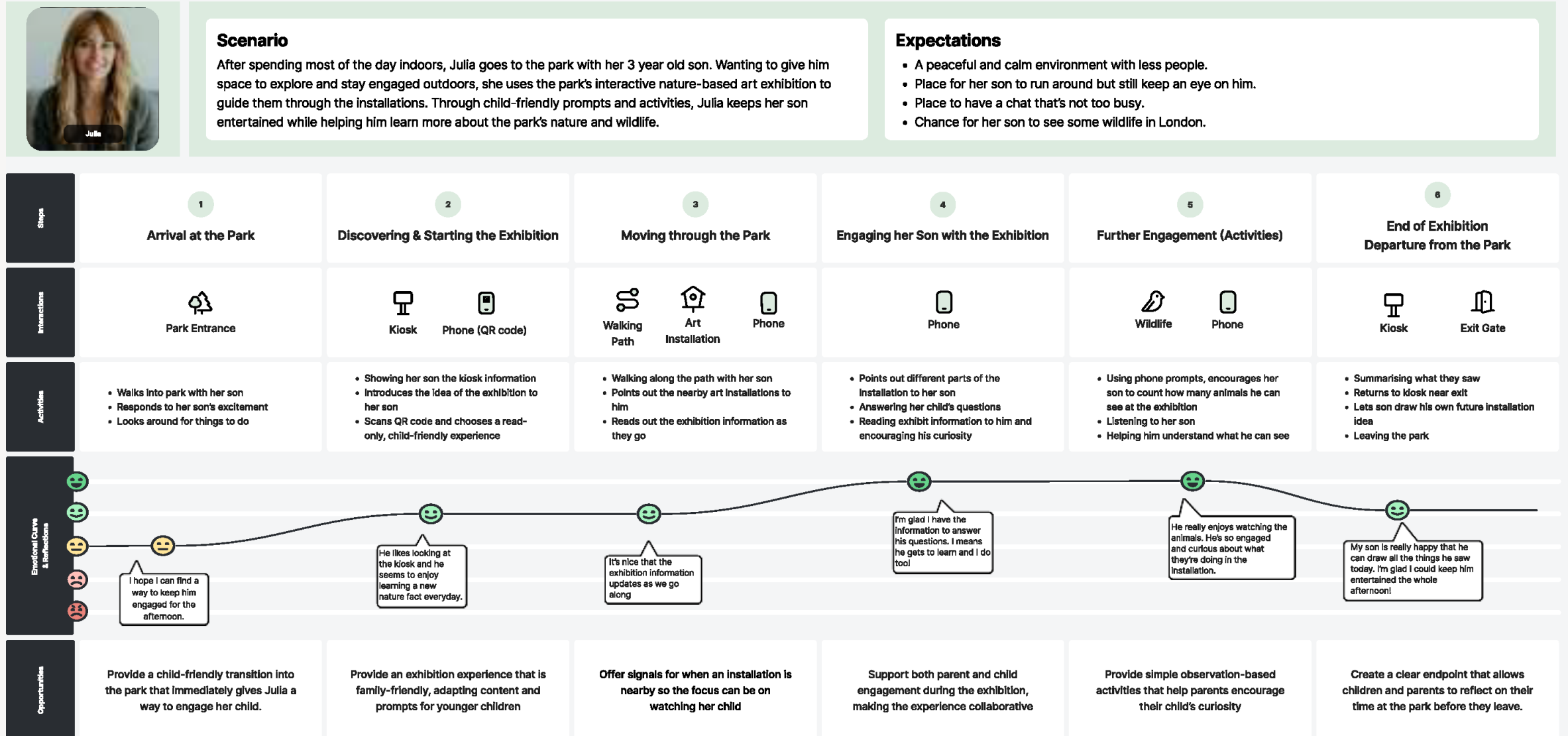
- Show how an experience could unfold with the new system
- Explore stages of interaction over time
- Help designers think about how a design changes the overall experience.

Current vs Future User Journey Maps

Current journey map	Future journey map
Describes the existing experience	Describes a possible future experience
Based on user research	Based on design ideas / concepts
Identifies problems and opportunities	Explores how the design could improve the experience

You can also use a current journey map as a **starting point / inspiration** for a future journey map—to explore how things might change (i.e. improve) as a result of the new design / system.

Example: Future User Journey Map



Questions?



Key concept
Mental models

Mental Models

- Mental models are representations of the external world that people construct in their minds to help them understand how things work and what to do in current and future situations
- Norman defines them as follows:

“People create mental models of themselves, others, the environment and the things with which they interact. These are conceptual models, formed through experience, training and instruction. These models serve as guides to help achieve our goals and in understanding the world.”

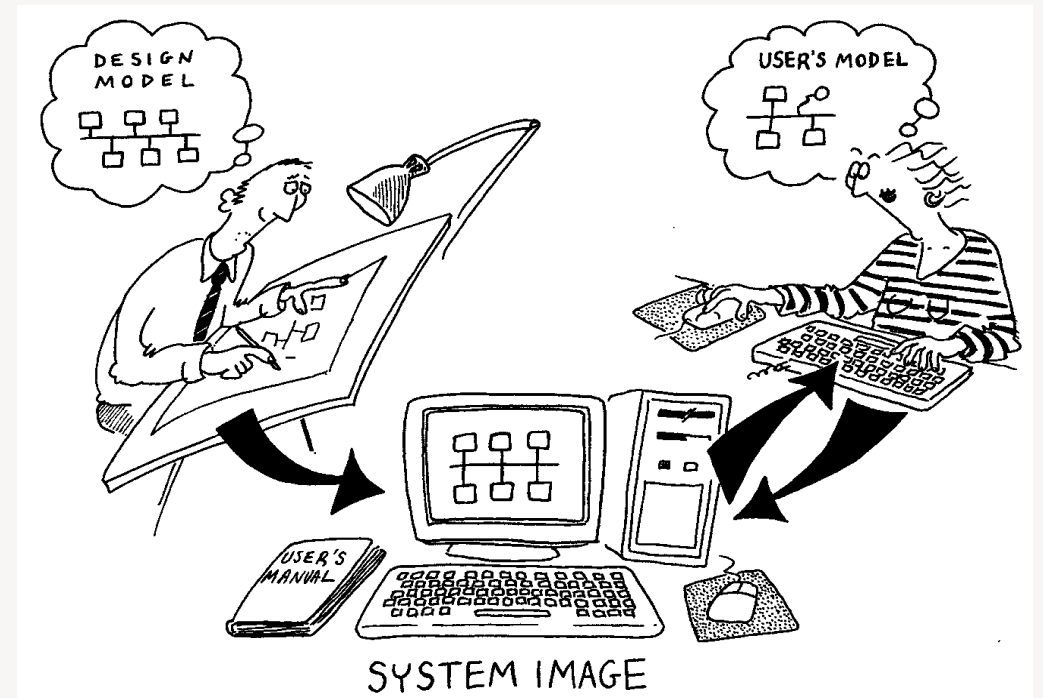
- Don Norman (2013) The Design of Everyday Things, revised and expanded edition.

Mental Models and Interaction Design

- Users need to understand how to interact with complex (and not so complex) digital technologies
- *“Most people have incomplete, unstable, easily confusable and superstitious mental models of systems and devices”* (Norman 1983)
- Interaction problems arise when a person’s mental model differs from the reality in ways that impact their interaction with a system
- Users may have a mental model of the task they want to do (particularly if it’s one they have done before with a different technology).
- But other users may have different mental models!
- Part of the challenge of designing good user interfaces and interaction, is to design for lots of users with different mental models.

Mental Models and Interaction Design (Don Norman)

- **Designer's model**
Designer's internal conception of the product.
- **System model/image**
What has been created that can be perceived (product + documentation...)
- **User's model**
User's conception (mental model) of the product, from interacting with the system image.



Questions?



Session 6 summary

- Conceptual design explores possible solutions before detailed design begins
- Designers generate ideas using rapid ideation techniques such as Crazy 8s
- Early concepts and ideas can be evaluated and refined using lightweight methods. e.g. dot voting, product triad, concept testing, design critiques...
- Designers then contextualise ideas, imagining how interactions unfold in real use situations, e.g. using scenarios, storyboards and future journey maps.
- Mental models describe users' understanding of how a system works, which design should support in both concepts and detailed interactions



GIVE TO GAIN

Empowering Future-Ready Professionals

INTERNATIONAL WOMEN'S DAY CELEBRATION



- **Keynote Talks**
- **Panel Session:** Collective Growth for Women in Science & Tech.
- **Mentoring Session:** on Next Steps in Career

WHEN?

9.30 – 16.30, 13th March

WHERE?

Ada Lovelace Room (A309),
College Building, City St.
George's, University of London,
Northampton Square, EC1V0HB

Scan for [Program Details](#)
and [Registration](#):

